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Homework 2

Sampling distributions form the foundation of statistical inference by describing the probability distribution of a sample statistic computed from random samples.

The normal distribution plays a fundamental role in sampling theory and serves as the foundation for many other sampling distributions. In a situation when a random sample is taken from a normal population with mean μ and variance σ^2 , then the resulting sample mean $\bar{X}$ follows an exact normal distribution, $\bar{X} \sim N(\mu, \sigma^2/n)$. In instances when the population distribution is not normal, the Central Limit Theorem implies that the sampling distribution of $\bar{X}$ is approximately normal for large sample sizes. This assumes the population has a finite mean and variance. Because of this property, the normal distribution forms the basis for both exact and asymptotic inference procedures and is closely connected to the t , chi-square, and F distributions.

The t distribution comes into play when we are making inference about a population mean but do not know the population variance. In the normal population setting, since the variance is unknown, we estimate it with the sample variance S^2 , which leads to the standardized statistic $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$, and this statistic follows a t distribution. Compared to the normal distribution, the t distribution places more probability in the tails, which reflects the added uncertainty from estimating the variance. As the sample size increases, the t distribution converges to the standard

normal distribution and is closely connected to the chi-square distribution through its dependence on S^2 .

The chi-square distribution comes up when we look at how the sample variance behaves from one sample to another. In the case of a normal population, if we scale the sample variance as $\frac{(n-1)S^2}{\sigma^2}$, this quantity follows a chi-square distribution with $n - 1$ degrees of freedom. This result depends on the assumption of normality and independence of the observations. The chi-square distribution is closely connected to the t and F distributions, since it forms the basis for the denominator of the t statistic and is used in constructing the F distribution.

The F distribution comes up when we want to compare the variability of two independent populations. If we draw random samples from two normal populations, the ratio of their sample variances, $F = \frac{S_1^2}{S_2^2}$, follows an F distribution with degrees of freedom based on the two sample sizes. This result depends on the assumptions of normality and independence for both samples. The F distribution is closely connected to the chi-square distribution, since it is constructed from the ratio of two scaled chi-square variables, and it plays an important role in procedures such as variance comparison and analysis of variance.